



Effects of Smoking on Outcomes of Hypertension

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Background



- Hypertension = approximately 47.3% prevalence in the US (AHA, 2021-2023 survey data)
- Affects over 1 billion adults worldwide, as it is a leading risk factor for CVD, kidney disease, and stroke
- Contributions to hypertension include lifestyle, diet, and physiological cofactors
 - Why? Hypertension can be asymptomatic (“silent killer”), meaning that it requires lifestyle management and medications.
- Smoking can increase hypertension risk as it damages artery walls, accelerates plaque buildup, and increases cardiovascular risks (heart attack, stroke).

Study Objectives



- **Question of interest:** How does smoking influence high blood pressure outcomes, and how do potential confounders and mediators change this effect when controlled for and on their own?
- Our main objective is to evaluate the effect of smoking on high blood pressure outcomes in middle-aged individuals (40-65 years) and to assess how this association changes after accounting for relevant confounders and mediators.
 - This age group is chosen because it presents the optimal connection between smoking exposure and high blood pressure.
 - Younger individuals may engage in smoking, but have not experienced the long-term hypertensive effects.

Study Design & Setting



- CDC's National Health and Nutrition Examination Survey (NHANES) 2021-2023 data
- Categories of data include: examination, dietary, laboratory, and questionnaire
- Study design includes:
 - Collating data according to SEQN (respondent sequence number) through laboratory visit and follow-up phone survey conducted in specialized Mobile Examination Centers (MECs)
 - **Main predictor of interest: Smoking**, categorized by two main variables:
 - SMQ020 - "Have you smoked at least 100 cigarettes in your life?"
 - SMQ040 - "Do you now smoke cigarettes?"
 - **Main outcome of interest: Hypertension**
 - BPXOSY (systolic) and BPXODI (diastolic); 3 measurements each and clinical definition for cutoffs of hypertension

Statistical Hypotheses



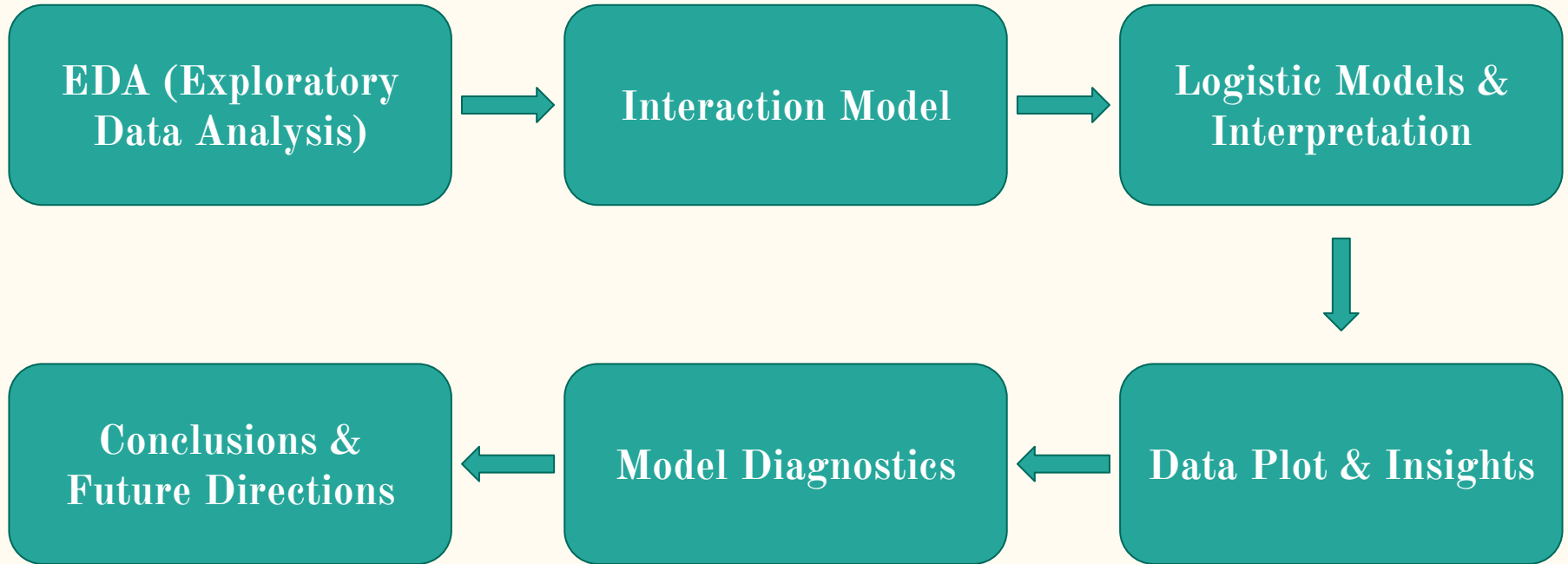
- **Primary statistical hypothesis:**

- H_0 : Smoking has no significant effect on high blood pressure outcomes ($\beta_{\text{smoking}} = 0$).
- H_A : Smoking has a significant effect on high blood pressure outcomes ($\beta_{\text{smoking}} \neq 0$).

- **Secondary statistical hypothesis:**

- H_0 : Controlling for confounders does not significantly change the effect of smoking on high blood pressure ($\beta_{\text{smoking}}^* = \beta_{\text{smoking}}$).
- H_A : Controlling for confounders significantly changes the effect of smoking on high blood pressure ($\beta_{\text{smoking}}^* \neq \beta_{\text{smoking}}$).

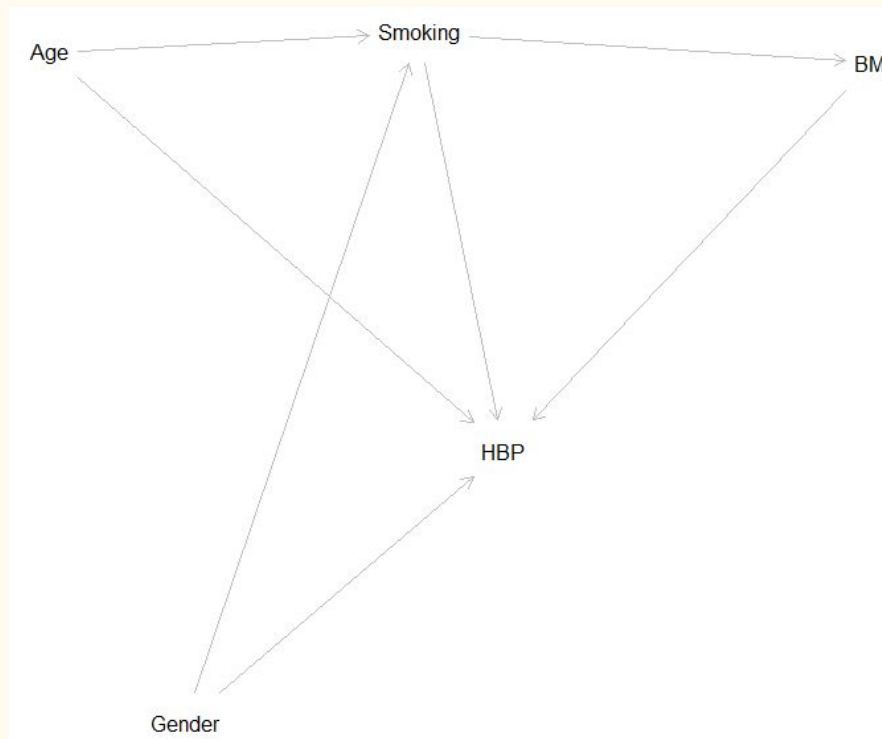
Data Analysis Workflow



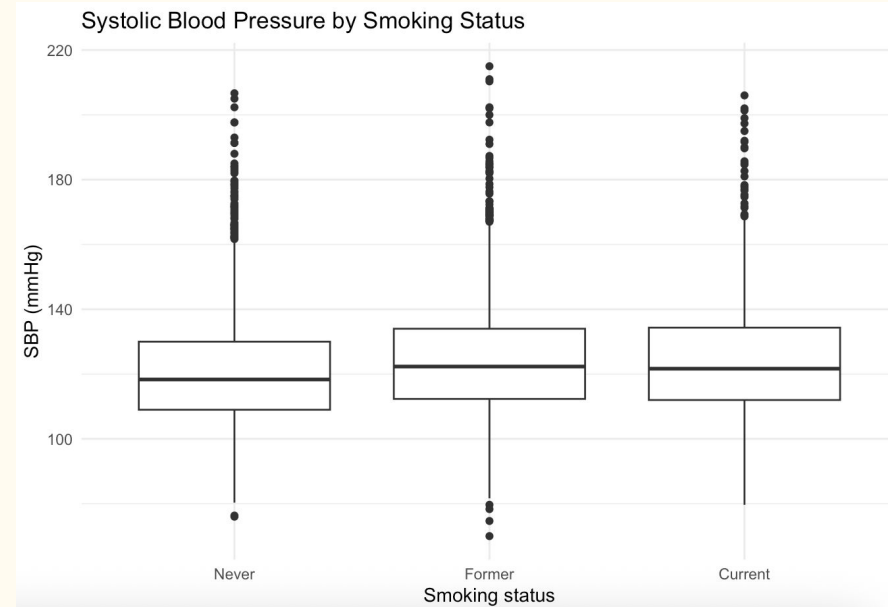
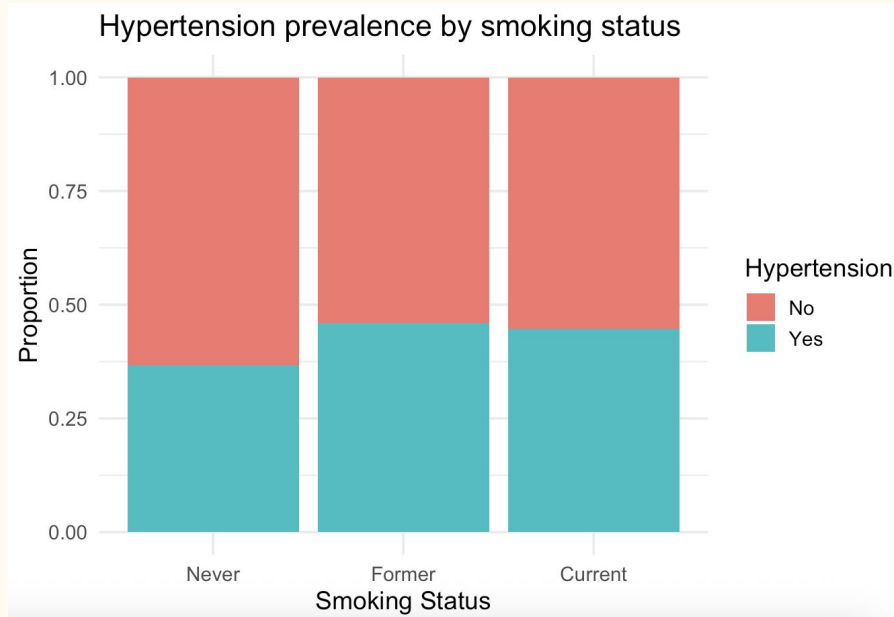


DAG of Variable Relationships

- Age and gender are possible confounders in the association between smoking and HBP (high blood pressure).
 - Smoking prevalence can differ by age group, or between men and women.
- BMI (body mass index) is a possible mediator.
 - Smoking may influence body weight/BMI and BMI influences blood pressure.



Exploratory Data Analysis



Takeaway: Both former and current smokers show higher prevalence of hypertension compared with never smokers. However, systolic blood pressure distributions are broadly similar across smoking groups, with median values around 120 mmHg.

Interaction Model



```
lm(formula = sbp ~ bmi * smoking + age + sex, data = dat2)
```

Residuals:

	Min	1Q	Median	3Q	Max
	-58.982	-10.407	-1.443	8.421	86.190

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	99.89365	1.24883	79.990	< 2e-16	***
bmi	0.13584	0.03694	3.677	0.000238	***
smokingFormer	0.60246	2.19050	0.275	0.783302	
smokingCurrent	9.11571	2.55275	3.571	0.000358	***
age	0.39593	0.01184	33.436	< 2e-16	***
sexFemale	-4.30330	0.42387	-10.152	< 2e-16	***
bmi:smokingFormer	-0.04844	0.07063	-0.686	0.492816	
bmi:smokingCurrent	-0.24562	0.08531	-2.879	0.004000	**

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 16.23 on 6044 degrees of freedom
(1401 observations deleted due to missingness)
Multiple R-squared: 0.1798, Adjusted R-squared: 0.1789
F-statistic: 189.3 on 7 and 6044 DF, p-value: < 2.2e-16

	2.5 %	97.5 %
(Intercept)	97.44550350	102.34179717
bmi	0.06341906	0.20826786
smokingFormer	-3.69171268	4.89662759
smokingCurrent	4.11141111	14.12001435
age	0.37271489	0.41914171
sexFemale	-5.13423777	-3.47236312
bmi:smokingFormer	-0.18690096	0.09001514
bmi:smokingCurrent	-0.41285083	-0.07838999

$$SBP = \beta_0 + \beta_1(BMI) + \beta_2(smoking) + \beta_3(BMI * smoking) + \beta_4(age) + \beta_5(sex) + \epsilon$$

- Former smoker: 95% CI of (-0.19, 0.09), $p = 0.49$ for the interaction term
- Current smoker: 95% CI of (-0.41, -0.08), $p < 0.01$ for the interaction term

Takeaway: There is not a significant interaction between BMI and former smoking status.

However, the interaction between BMI and current smoking was statistically significant, with a negative coefficient of -0.25.

Logistic Models



Logistic model without smoking:

```
glm(formula = hypertension ~ bmi + age + sex, family = binomial(),
     data = dat2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-4.028071	0.133402	-30.195	< 2e-16 ***
bmi	0.051892	0.003728	13.921	< 2e-16 ***
age	0.040232	0.001386	29.018	< 2e-16 ***
sexFemale	-0.261783	0.054583	-4.796	1.62e-06 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 9480.5 on 7452 degrees of freedom
Residual deviance: 8053.6 on 7449 degrees of freedom
AIC: 8061.6

Number of Fisher Scoring iterations: 4

Logistic model with smoking:

```
glm(formula = hypertension ~ smoking + age + sex + bmi, family = binomial(),
     data = dat2)
```

Coefficients:

	Estimate	Std. Error	z value	Pr(> z)
(Intercept)	-3.314337	0.155030	-21.379	< 2e-16 ***
smokingFormer	0.058395	0.065837	0.887	0.37509
smokingCurrent	0.275950	0.079524	3.470	0.00052 ***
age	0.030981	0.001647	18.810	< 2e-16 ***
sexFemale	-0.234264	0.055745	-4.202	2.64e-05 ***
bmi	0.044664	0.003888	11.487	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

Null deviance: 8158.3 on 6051 degrees of freedom
Residual deviance: 7584.4 on 6046 degrees of freedom
(1401 observations deleted due to missingness)
AIC: 7596.4

Number of Fisher Scoring iterations: 4

Takeaway: BMI, age, and sex remain significant predictors in both models. Current smoking shows a statistically significant association with hypertension

Logistic Models Interpretation



Logistic model without smoking:

Variable	Estimate	OR	95% CI
BMI	0.05	1.05	(1.046, 1.061)
Age	0.04	1.04	(1.038, 1.044)
Sex	-0.26	0.77	(0.692, 0.857)

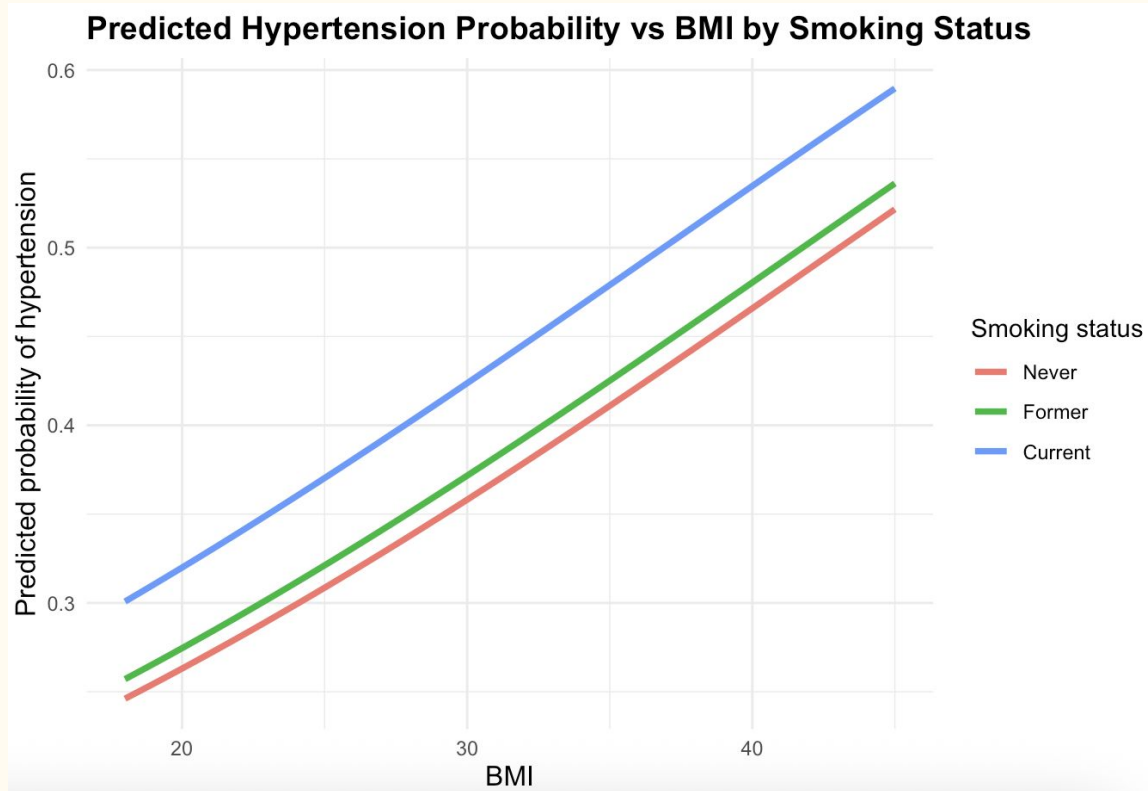
Each one-unit increase in BMI is associated with roughly a 5% increase in the odds of hypertension

Logistic model with smoking:

Variable	Estimate	OR	95% CI
BMI	0.045	1.05	(1.038, 1.054)
Age	0.031	1.03	(1.028, 1.035)
Sex	-0.234	0.79	(0.709, 0.882)
Smoking Former	0.058	1.06	(0.932, 1.206)
Smoking Current	0.276	1.32	(1.127, 1.54)

Takeaway: Current smokers have an odds ratio of 1.32, meaning they have about 32% higher odds of hypertension compared with never smokers. Former smokers do not show a statistically significant association with hypertension.

Predicted probability of hypertension across BMI levels for different smoking groups



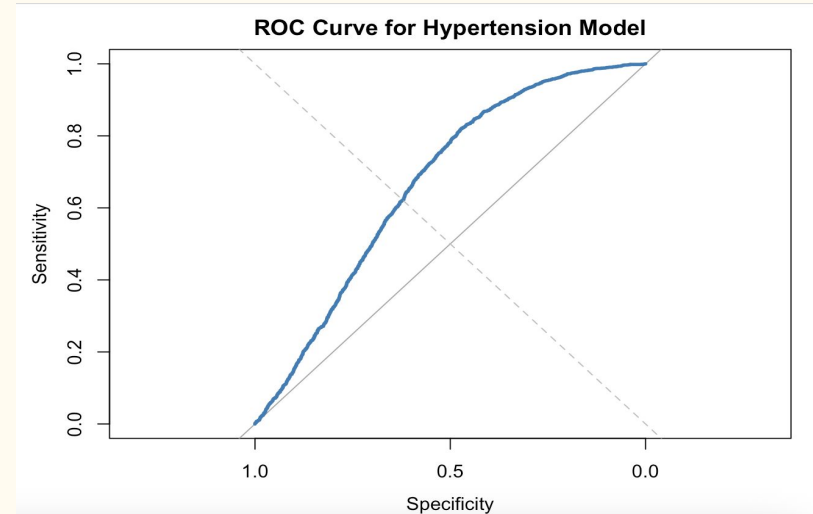
- ❑ Predicted probability of hypertension increases steadily with BMI across all smoking groups. BMI is a strong metabolic risk factor
- ❑ Current smokers consistently show higher predicted risk compared with never smokers.

Model Diagnostics:

	GVIF	Df	$GVIF^{1/(2 \cdot Df)}$
smoking	1.063892	2	1.015604
age	1.049296	1	1.024352
sex	1.022592	1	1.011233
bmi	1.015688	1	1.007813



- ❑ **Multicollinearity:** All VIFs were close to 1, indicating there is no evidence of problematic correlation between predictors
- ❑ **Influential Observation:** A small number of observations exceeded the Cook's distance threshold $4/n$, but no single observation showed disproportionately large influence on the model estimates
- ❑ **AUC:** 0.67, if we randomly pick one person with hypertension and one without, the model will correctly rank their risk about 67% of the time. This is expected given the limited number of predictors included in the model.



Discussion



Former smokers **did not** show significantly higher hypertension risk compared with never smokers after adjustment.

Current smokers showed higher odds of hypertension compared with never smokers, suggesting **smoking is associated with increased cardiovascular risk.**

Smoking may influence hypertension **through multiple pathways, including metabolic and cardiovascular mechanisms.**

Overall, traditional metabolic risk factors such as **BMI and age appear to have stronger associations with hypertension** than smoking status in this dataset.



Limitation & Future Directions

- **Unequal sample sizes across smoking groups:** 3640 non, 1539 former, 873 current; may affect precision of estimates within groups (smaller n, potentially less precise)
- **Cross-Sectional design:** Causal relationship between smoking and hypertension cannot be established using cross sectional dataset.
- *****Smoking status is self-reported in questionnaires!*****
- **Future questions:**
 - How do results of our study hold or contrast in other populations (outside of the US)?
 - **Longitudinal Analysis:** Future study using cohort data can better evaluate the temporal relationship between smoking and hypertension



Thank you!

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Final Model v2 (BMI as a mediator)



```
glm(formula = hypertension ~ age + sex + smoking, family = binomial(), data = dat2)
```

COEFFICIENTS	ESTIMATE	STD. ERROR	Z VALUE	PR(> Z)
(Intercept)	-1.981333	0.096331	-20.568	< 2e-16 ***
age	0.030422	0.001615	18.836	< 2e-16 ***
sexFemale	-0.185265	0.054947	-3.372	0.000747 ***
smokingFormer	0.091167	0.065191	1.398	0.161975
smokingCurrent	0.244229	0.078722	3.102	0.001919 **



Odds Ratios for Model v2

	TERM	OR	CI [LCL, UCL]
(Intercept)	(Intercept)	0.14	[0.114, 0.166]
age	age	1.03	[1.028, 1.034]
sexFemale	sexFemale	0.83	[0.746, 0.925]
smokingFormer	smokingFormer	1.10	[0.964, 1.245]
smokingCurrent	smokingCurrent	1.28	[1.094, 1.489]

BMI as mediator model: current smokers $OR = 1.28^{**}$, former smokers $OR = 1.10$; results consistent with BMI excluded from model, with former smoker still not being significant, though elevated by 4% compared to confounder model